

Why Agents?

The Case for Autonomous AI

By the end of this session, you'll identify chatbot gaps, classify business problems into AI patterns, and quantify the cost of under-investing in agentic AI.



Observation



Reflection



Identify Gaps

Spot 5 critical limitations in current chatbot workflows



Classify Patterns

Match real-world business problems to AI patterns

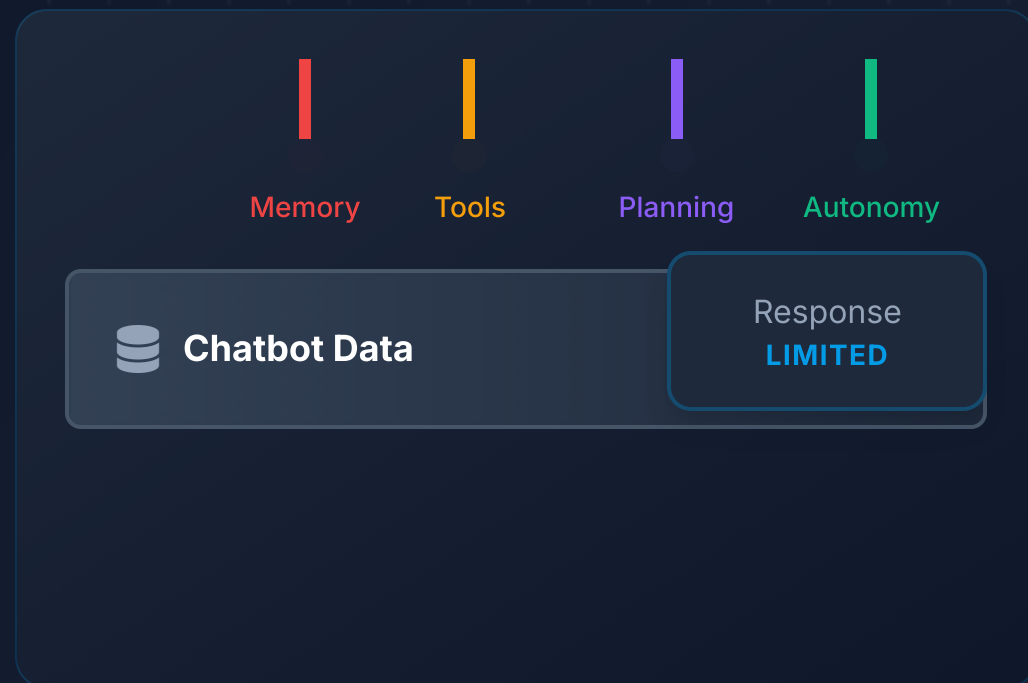


Quantify ROI

Measure the tangible cost of agentic AI investment

The Problem — Why This Matters Now

Your chatbot is leaking business value through five critical gaps. Here's why the status quo is costing you money.



Fragmented tasks stall without follow-through

Chatbots can only handle one instruction at a time, leaving workflows incomplete



No memory causes repeated context loss

Each session starts fresh, forcing users to re-explain their situation



Single replies can't drive outcomes

No planning capability means no multi-step execution



Humans babysit tools, wasting time

Constant supervision required for every interaction

73%

Of enterprises report chatbot limitations

\$2.4M

Average annual cost of manual workflows

4.2h

Average time saved with agentic AI

Learning Journey — From Gaps to Patterns

A visual roadmap through the 6 key sub-topics that will transform your understanding of autonomous AI



The 5 Gaps

Memory, Tools, Planning



Agent Loop

Goal → Plan → Act



Maturity Model

Level 0 → Level 3



Decision Framework

Choose the right pattern



Business Cost

ROI of agentic AI



Architecture

5 patterns to use

● Start: Chatbot Limitations

End: Agent Patterns ●



Identify Gaps

Learn the 5 critical limitations of current chatbot workflows



Classify Patterns

Match real-world business problems to AI patterns

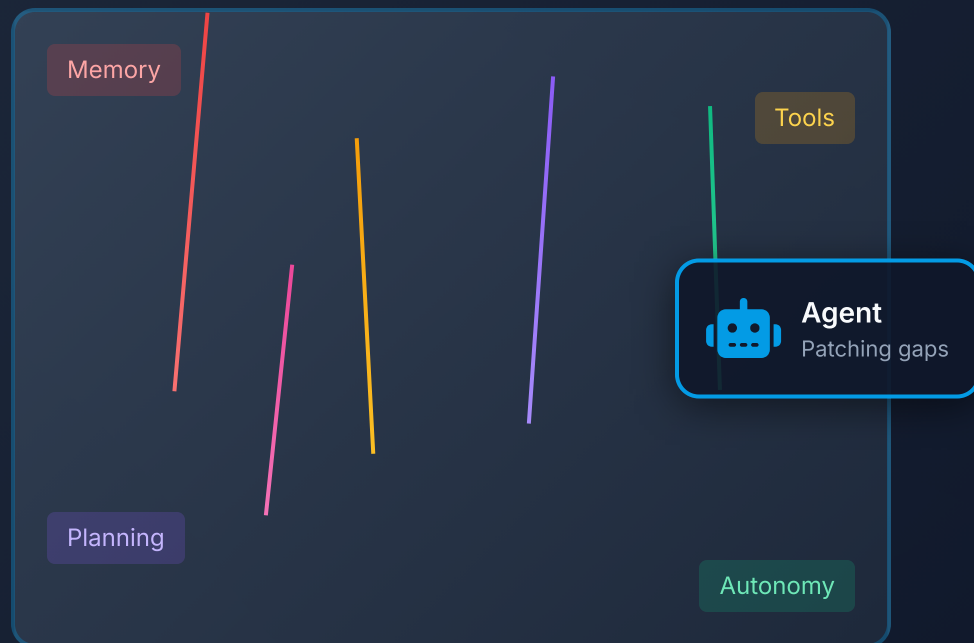


Quantify ROI

Measure the tangible cost of agentic AI investment

The 5 Gaps That Agents Close

What chatbots cannot do — and how autonomous AI agents fix each limitation



Memory Gap

CRITICAL

Chatbot forgets everything after session ends — agent persists context across time



Tools Gap

HIGH

Chatbot can only produce text — agent calls APIs, reads files, writes to systems



Planning Gap

MEDIUM

Chatbot answers one question — agent decomposes goals into ordered steps



Autonomy Gap

HIGH

Chatbot waits for human input — agent continues until goal is complete



Multi-Step Reasoning

CRITICAL

Chatbot handles one instruction — agent manages chains of dependent decisions

In Practice — Competitor Analysis Request

Business Scenario: Sales Manager Needs Report

A sales manager asks for a competitor analysis report — see which gaps cause a chatbot to fail and how an agent closes each one.



Chatbot Approach

Fails on 4 out of 5 gaps

✖

Failed

1

Receives request
Understands the task

✔

Success

2

Memory gap
Forgets previous interactions

✖

Failed

3

Tools gap
Cannot access competitor data

✖

Failed

4

Planning gap
No multi-step execution

✖

Failed

5

Autonomy gap
Waits for human input

✖

Failed

⚠

Result: Incomplete report, no follow-through



Agent Approach

Closes all 5 gaps

✔

Success

1

Receives request
Understands the goal

✔

Success

2

Memory persistence
Maintains context across sessions

✔

Success

3

Tool integration
Calls APIs, searches web, reads files

✔

Success

4

Planning & execution
Decomposes goal into steps

✔

Success

5

Autonomous completion
Continues until goal achieved

✔

Success

✔

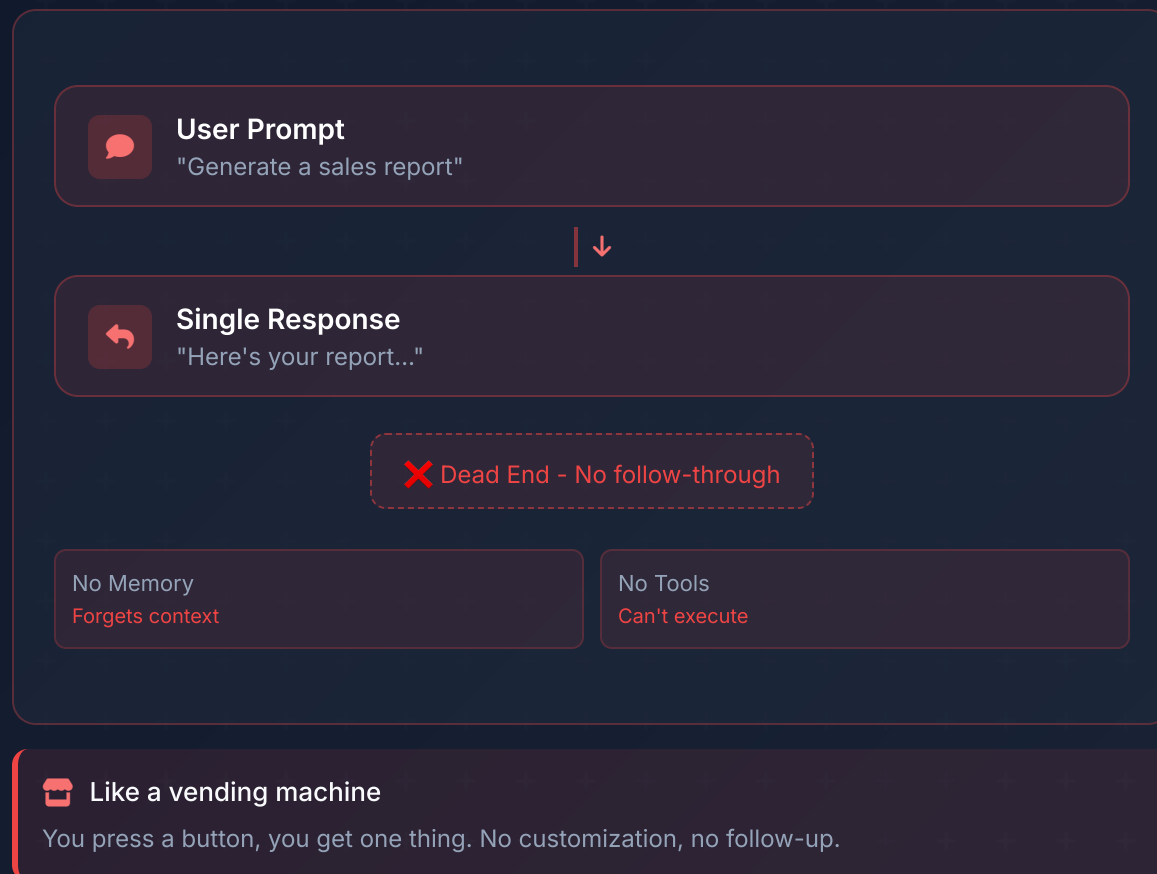
Result: Complete report with follow-up



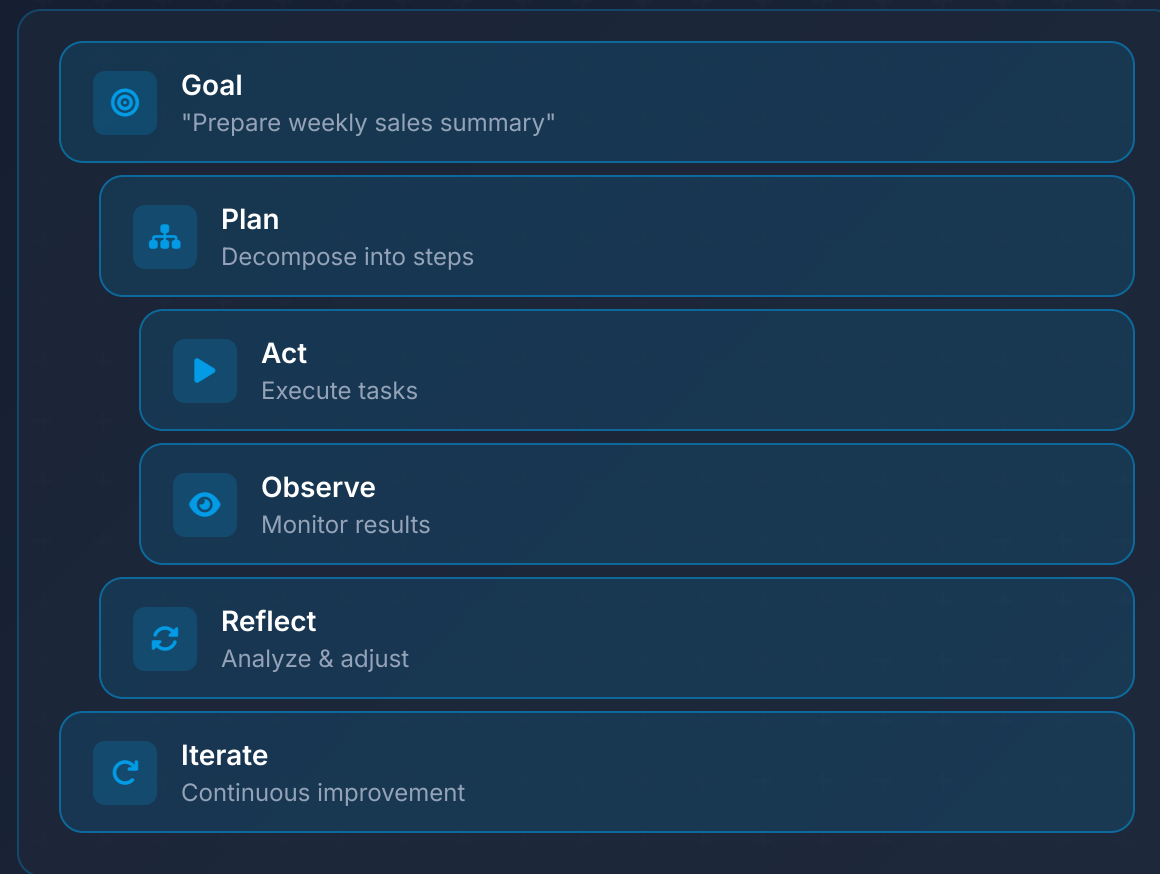
From Prompt → Response to Goal → Plan → Act → Reflect → Iterate

Understanding the fundamental shift from reactive chatbots to proactive autonomous agents

Chatbot Pattern Vending Machine Analogy



Agent Pattern Project Manager Analogy



• Key Insight: Agents transform single-turn interactions into continuous, goal-oriented workflows

Like a project manager

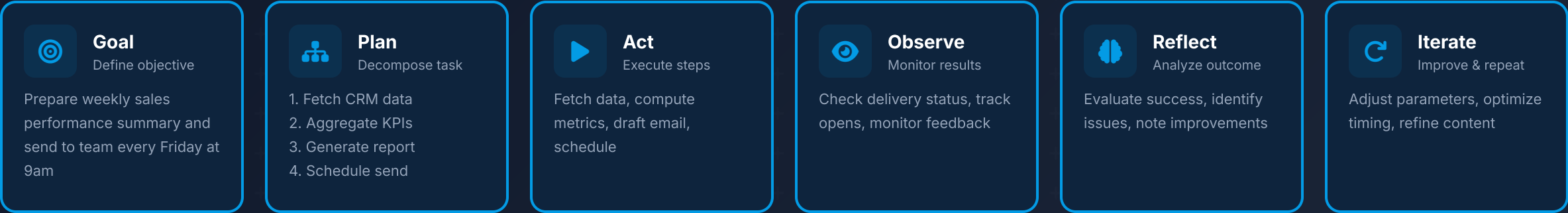
You give a goal, they plan steps, execute, check results, and adjust.

Chatbot: 1 interaction → Agent: ∞ iterations

Walkthrough — Weekly Sales Summary, Fridays 9am

Task: Prepare & send weekly report

A complete example showing how an agent handles a recurring task through the full loop: Goal → Plan → Act → Observe → Reflect → Iterate



Detailed Process
Step-by-step execution

Status: Running

1

Fetch CRM data
Query Salesforce API for last week's sales data

2 min

2

Aggregate KPIs
Calculate total revenue, deals closed, conversion rate

1 min

3

Generate report
Create formatted summary with charts and analysis

3 min

4

Schedule send
Set up automated email for Friday 9am

1 min

Human Checkpoint
Where do you want oversight?

Thought Prompt:

Think of a recurring task in your role. Which steps could an agent handle autonomously — and where would you want a human checkpoint?

Agent handles:

Data fetching

Report generation

Scheduling

Human reviews:

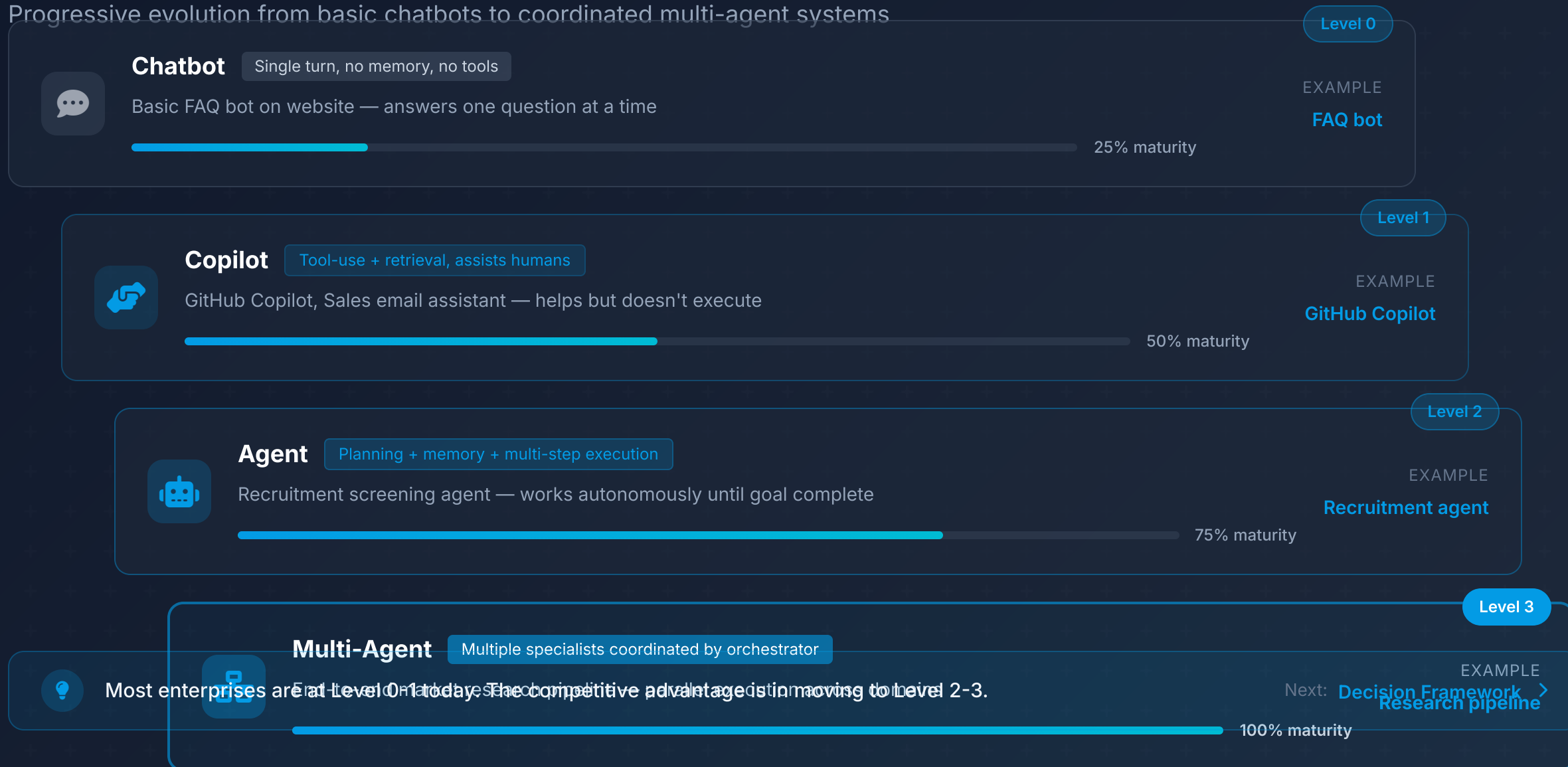
Final approval

Data validation

Exception handling

Agent Maturity Model — Level 0 → Level 3

Progressive evolution from basic chatbots to coordinated multi-agent systems



Challenge — Where Are You Today?

Assessment: **Maturity Level Evaluation**

Map your organization to the maturity model and identify steps to advance one level.



Maturity Levels

Current state assessment

4 Levels

Level 0: Chatbot

Example: FAQ bot on website

Single turn, no memory

Level 1: Copilot

Example: GitHub Copilot, Sales email assistant

Tool-use + retrieval

Level 2: Agent

Example: Recruitment screening agent

Planning + memory + multi-step

Level 3: Multi-Agent

Example: End-to-end market research pipeline

Coordinated specialists



Self-Assessment

Evaluate your current position

Think

1 Map teams to maturity levels

Which departments use which AI patterns?

2 Identify blockers

Data, tools, risk, or resource constraints?

3 Define next-level pilot

What would moving one level up look like?

Thought Challenge

Where does your organization sit on this model today? What would it take to move one level up?

Decision Framework: Chatbot vs RAG vs Agent vs Multi-Agent

1

Does the answer exist in a fixed document or database?

YES → RAG

NO → Continue

2

Does the task require a single response with no follow-up action?

YES → Chatbot

NO → Continue

3

Does the task require multiple steps, memory, or tool use?

YES → Agent

NO → Continue

4

Does the task require multiple specialized capabilities working in parallel?

RAG

Document-based answers

Multi-Agent

Single-turn responses

Chatbot

Single Agent

Agent

Multi-step execution

Multi-Agent

Parallel specialists

 **Pattern Comparison**

Key characteristics

RAG	Document retrieval
Chatbot	Single response
Agent	Multi-step tasks
Multi-Agent	Parallel execution

 **Key Insight**

Most enterprises are at Level 0–1 today. The competitive advantage is in moving to Level 2–3.

Apply the Framework — 4 Scenarios

Pattern Selection: **RAG vs Chatbot vs Agent vs Multi-Agent**

Map 4 real business scenarios to the recommended AI pattern using the decision framework.



Answer customer questions about return policy RAG

Fixed knowledge in documents

✓ Uses document retrieval for accurate answers



Generate one-time product description Chatbot

Single response, no follow-up needed

✓ Quick content generation without persistence



Screen 200 resumes and schedule interviews Agent

Multiple steps + tools + memory

✓ Autonomous workflow with tool integration



Run competitive intelligence report Multi-Agent

Parallel specialties across systems

✓ Coordinated analysis across multiple data sources



EdYoda Agent Builder

Integration with n8n orchestration

User Input

Business problem description

→

Pattern Analysis

Decision framework applied

Agent Selection

Appropriate AI pattern chosen

→

Execution

n8n workflow triggered



Key Insight

Use EdYoda Agent Builder for pattern selection, then pair with n8n for complex orchestration workflows.



Real Business Cost of Not Using Agents

Three critical failure scenarios with measurable business impact



Missed Context
Support Agent Failure

● Customer Support

Agent gives wrong advice due to no memory of prior interactions

Customer Churn Rate

+15%

Business Impact

HIGH

- Customer satisfaction drops
- Support tickets escalate
- Revenue loss: \$50K-100K



No Follow-Through
Deal Stagnation

● Sales Process

Chatbot drafts proposal but cannot send or track it

Deal Velocity

-30%

Business Impact

MEDIUM

- Deals go cold
- Sales cycle extends
- Revenue loss: \$25K-50K



Zero Persistence
Compliance Risk

● Data Management

Session times out, all context is lost

Compliance Risk

CRITICAL

Business Impact

CRITICAL

- Audit failures
- Legal penalties
- Revenue loss: \$100K+

⚠ Total estimated cost of not using agents: **\$175K-250K annually**

Based on industry averages for mid-size enterprise

ROI Framing — Make the Cost Visible

Quantify baseline costs, estimate uplift by pattern level, and compute ROI to justify agent investment.

Cost Analysis: Hidden Business Impact



Cost Funnel Analysis

Input → Process → Output

Input Cost

\$50K/month

High

Processing

\$35K/month

Med

Output Value

\$120K/month

Low

Time Saved

45%

Error Reduction

78%



ROI Calculator

Investment vs Returns



Month	Current Cost (\$K/month)	Projected Cost (\$K/month)
Month 1	85	85
Month 2	85	75
Month 3	85	65
Month 4	85	55
Month 5	85	45
Month 6	85	45

Current Cost

\$85K/month

Projected Savings

\$40K/month



Thought Prompt

Which failure have you witnessed?



Missed Context

Support agent gives wrong advice due to no memory



No Follow-Through

Chatbot drafts proposal but cannot send it



Zero Persistence

Session times out, context is lost



Agentic Architecture Patterns & When to Use Each

Five proven patterns for building autonomous AI systems — choose the right approach for your use case



Single Agent

Self-contained

One LLM + tools + memory for sequential tasks

When to use:

Task is self-contained, sequential, and doesn't require parallel processing

Example: [Email drafting agent](#)



RAG Agent

Document-based

Single agent with knowledge base retrieval

When to use:

Task requires grounding in documents or proprietary data

Example: [Support FAQ bot](#)



Tool-Calling Agent

API Integration

Agent with external API integrations

When to use:

Task requires reading from or writing to external systems

Example: [CRM data sync](#)



Orchestrator-Worker

Multi-agent

One planning agent coordinates multiple specialists

When to use:

Task has parallel workstreams requiring different expertise

Example: [Market research pipeline](#)



Human-in-the-Loop

High-stakes

Agent pauses at checkpoints for human approval

When to use:

Task involves high-stakes decisions or regulated actions

Example: [Financial approvals](#)



Next: Prompt Engineering

Learn how to communicate with agents precisely enough to make them reliable

Now that you can identify patterns, master the language of agent communication

AI Pattern Advisor — Built This Session

Intelligent recommendation system for matching business problems to optimal AI patterns



AI Pattern Advisor

Version 1.0

Purpose

Recommends optimal AI pattern for any business problem with maturity level assessment

Inputs

Free-text problem description

Outputs

Pattern + level + rationale

Active

Key Capabilities

- Pattern classification
- Maturity assessment
- Risk identification
- Justification generation

This agent was built during today's session to demonstrate pattern-matching capabilities

Architecture Flow

Input → Processing → Output

User Input
Problem description

System Prompt
Role: AI architect

Decision Logic
4-question framework

LLM
Pattern analysis

Structured Output

Pattern + level + rationale + risk flag

Pattern RAG	Level Level 2	Justification 3 lines	Risk Medium
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Knowledge Base

3 sources

4-Pattern Framework
Decision tree logic

Maturity Model
Level descriptions

12 Scenarios
Real-world examples

Next: Recap & What's Next →

Recap + What's Next

Key concepts learned and next steps in your AI journey



5 Gaps Chatbots Leave Open

Memory, Tools, Planning, Autonomy, Multi-step Reasoning



Agent Loop: Goal to Iteration

Plan → Act → Observe → Reflect → Iterate



4 Levels of Agent Maturity

Level 0 (Chatbot) → Level 3 (Multi-Agent)



Decision Framework: Right Pattern Every Time

Chatbot vs RAG vs Agent vs Multi-Agent



Business Cost of Staying Chatbots

Missed context, no follow-through, zero persistence



5 Agentic Architecture Patterns

Single Agent, RAG, Tool-Calling, Orchestrator, Human-in-Loop

→ Bridge to Next Topic

You can now identify the right AI pattern for any business problem. Next — you'll learn how to talk to agents precisely enough to make them

💡 Reflection Prompt

Pick one process in your organization currently handled by a chatbot or manual workflow. Which agent pattern would you use to upgrade it —